

LUKE OFFSHORE BUOY 42040, USA

WMO: 995000

Lat: 29.208N

Lon: 88.226W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (W06)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	8.3	-1.1	3.4	9.5	0.5	3.9	10.4	16.2	12.1	15.2	13.4	9.2	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.3	30.8	25.9	30.5	25.7	30.1	25.5	27.2	29.8	26.8	29.6	26.3	29.3	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.0	29.3	25.8	21.1	29.0	25.4	20.6	28.8	85.7	29.8	83.5	29.7	81.7	29.4	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	13.0	11.1	DB	3.8	31.9	2.0	1.0	2.4	32.6	1.2	33.2	0.0	33.8	-1.4	34.6	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	16.1	16.4	19.0	21.3	24.4	27.5	28.3	28.6	27.3	24.2	20.4	18.0
	DBStd	5.17	4.07	3.36	2.96	2.17	1.90	1.29	1.16	1.26	1.18	2.35	3.30	3.76
	HDD10.0	9	6	2	0	0	0	0	0	0	0	0	0	1
	HDD18.3	265	91	67	28	5	0	0	0	0	0	2	19	53
	CDD10.0	4628	194	182	279	338	447	524	566	576	520	439	313	249
	CDD18.3	1843	21	14	49	93	188	274	308	317	270	182	82	43
	CDH23.3	16540	1	0	7	98	1140	3073	3826	4048	2984	1143	189	32
CDH26.7	4708	0	0	0	0	37	826	1387	1617	767	74	0	0	
Wind	WSAvg	5.9	7.2	6.8	6.3	6.3	5.2	4.6	4.4	4.5	5.5	6.4	7.0	7.2
Precipitation	PrecAvg	1469	120	96	122	123	98	129	165	166	149	104	89	107
	PrecMax	2160	374	213	333	462	261	403	261	312	384	297	366	272
	PrecMin	894	3	18	16	13	11	27	75	64	45	19	9	27
	PrecStd	265	85	52	68	90	63	75	52	65	74	72	72	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.5	22.9	24.1	25.7	28.1	30.8	31.3	31.8	30.2	28.5	26.2	24.6
		MCWB	22.7	22.1	21.6	23.8	24.1	25.5	26.0	26.2	24.8	24.9	24.1	22.9
	2%	DB	22.9	22.1	23.5	25.0	27.6	30.2	30.7	31.1	29.7	28.0	25.6	24.2
		MCWB	21.9	20.9	21.3	23.1	24.1	25.3	25.8	26.0	24.7	24.8	23.0	22.3
	5%	DB	22.3	21.5	23.0	24.4	27.0	29.7	30.2	30.7	29.2	27.7	25.0	23.7
		MCWB	20.9	20.1	21.1	22.5	23.8	25.2	25.6	25.9	24.7	24.7	22.1	21.8
	10%	DB	21.6	20.7	22.5	23.9	26.7	29.3	30.0	30.3	29.0	27.0	24.5	23.0
		MCWB	19.9	19.2	20.7	21.9	23.7	25.0	25.4	25.7	24.7	23.9	21.3	21.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.5	22.7	24.7	25.7	27.0	27.4	28.0	26.9	27.3	25.2	23.5
		MCDB	23.1	22.7	23.3	25.4	27.0	28.9	30.2	30.7	28.4	27.5	25.9	24.1
	2%	WB	22.5	21.4	22.2	23.9	25.2	26.4	26.9	27.4	26.2	26.2	24.0	23.0
		MCDB	22.8	21.8	23.0	24.9	26.6	29.0	29.9	30.4	28.5	27.3	25.3	23.7
	5%	WB	21.3	20.5	21.7	23.2	24.8	26.0	26.5	26.9	25.8	25.6	22.8	22.5
		MCDB	22.1	21.1	22.7	24.2	26.4	28.8	29.7	30.0	28.4	27.1	24.7	23.5
	10%	WB	20.2	19.7	21.2	22.7	24.3	25.6	26.2	26.5	25.5	24.7	21.8	21.4
		MCDB	21.5	20.6	22.3	23.8	26.2	28.6	29.5	29.8	28.3	26.7	24.2	22.9
Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	2.7	2.3	1.8	2.1	2.2	2.3	2.0	2.2	3.2	3.7
		MCDBR	3.2	3.9	2.3	2.3	1.9	2.4	2.3	2.5	2.1	2.3	3.0	3.7
		MCWBR	4.1	4.9	2.8	2.7	1.9	2.1	2.0	2.2	2.3	3.1	3.8	4.7
	5% WB	MCDBR	3.6	4.3	2.5	2.1	1.8	2.4	2.2	2.5	2.1	2.7	3.7	4.0
		MCWBR	4.4	5.1	3.0	2.9	2.2	2.3	2.2	2.3	2.4	3.6	5.0	5.5
Clear-Sky Solar Irradiance	taub	0.353	0.367	0.379	0.412	0.436	0.453	0.479	0.477	0.439	0.395	0.366	0.357	
	taud	2.524	2.493	2.452	2.351	2.326	2.328	2.274	2.287	2.352	2.468	2.491	2.523	
	Ebn at Noon	884	900	909	885	860	840	817	817	838	859	860	861	
	Edh at Noon	89	100	110	126	130	129	135	133	120	101	92	86	
All-Sky Solar Radiation	RadAvg	3.12	3.83	5.12	6.13	6.90	6.53	6.03	5.59	5.25	4.61	3.55	2.80	
	RadStd	0.27	0.51	0.43	0.57	0.34	0.49	0.39	0.44	0.33	0.39	0.35	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+132	+85

Nomenclature: See separate page